

Nirvan Bhattacharyya, PhD
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EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN 2018 – 2023

PhD in Chemical Engineering – Advisor: Lea Hildebrandt Ruiz

Dissertation titled: Fate and Oxidative Impact of Chlorine in Indoor and Outdoor Air

UNIVERSITY OF MICHIGAN, ANN ARBOR 2014 – 2018

Bachelor of Science in Chemical Engineering

RESEARCH EXPERIENCE

GOETHE UNIVERSITY FRANKFURT, Institute for Atmospheric and Environmental Science

Postdoctoral Research Scientist – Curtius Group 2025 – Present

- Operated and maintained gas and particle measurements at Taunus Observatory field station.
- Prepared, quality assured, and submitted Taunus Observatory measurement data to the ACTRIS network.
- Designed and executed flow tube and mass spectrometer measurements at the CLOUD chamber at CERN.
- Mentored and advised bachelor's, master's, and doctoral students, including paper and thesis review.

CARNEGIE MELLON UNIVERSITY, Center for Atmospheric Particle Studies

Postdoctoral Research Associate – Donahue Group 2023 – 2025

- Deployed gas and particle phase mass spectrometer at CLOUD measurement campaign at CERN
- Designed and conducted experiments on atmospheric particle formation and partitioning.
- Trained and supported PhD and Master's students with mass spectrometer operation and analysis.
- Analyzed multi-instrumental data to assess isoprene-derived particle growth in a new volatility-growth framework. Paper published in *Environmental Science and Technology Air* (2025)

UNIVERSITY OF TEXAS AT AUSTIN, Center for Energy and Environmental Resources

Graduate Research Assistant – Hildebrandt Ruiz Group 2018 – 2023

- Demonstrated impact of chlorine and hydroxyl radicals on m-xylene oxidation and formation of gas and particle phase products. Paper published in *Environmental Science: Atmospheres* (2023).
- Assessed chlorine oxidative flow reactor for studying atmospheric oxidation in field deployments and laboratory studies with Aerodyne Research, Inc. Papers in preparation.
- Collaborated with Drs. Novoselac and Misztal to assess bleach interaction with masks. Found masks act as a long-term source. Published in *Environmental Science and Technology* (2023).
- Advised an undergraduate researcher in mass spectrometer data analysis of indoor cleaning.

UNIVERSITY OF MICHIGAN, ANN ARBOR, Civil and Environmental Engineering

Research Assistant – Demond Group 2015 – 2017

- Studied partitioning and solubility of pesticides. Published in *Journal of Environmental Engineering* (2017).

TEACHING AND ADVISORY EXPERIENCE

GOETHE UNIVERSITY FRANKFURT, Department of Geosciences

Doctoral Mentor – Eva Sommer (Uni-Vienna Physics)

Winter – Summer 2026

Master's Thesis Supervisor – Leo Künstler

Winter 2026

Bachelor's Thesis Supervisor – Christoph Schmidt

Winter 2026

Tutorial Instructor – Physics and Chemistry of the Atmosphere II

Winter 2026

UNIVERSITY OF TEXAS AT AUSTIN, Department of Chemical Engineering

Undergraduate Research Mentor – Priya Manchiraju

Fall – Spring 2021/2022

Teaching Assistant – Intro to Chemical Engineering Analysis

Fall 2020

Teaching Assistant – Process and Project Laboratory

Fall 2019

PROFESSIONAL MEMBERSHIPS & SERVICE

Member of the CLOUD Collaboration (2023 to present)

Co-organizer for CLOUD-DOC Final Conference and CLOUD18 Data Workshop, March 2026

Member of American Association of Aerosol Researchers (AAAR, 2020 to present)

Member of American Geophysical Union (AGU, 2022 to present)

Session Convener and Chair, *Aerosols, New Particle Formation and Clouds*, AGU25, Dec. 2025

Member of American Institute of Chemical Engineering (AIChE, 2015 to present)

Peer Reviewer for *Environmental Science: Atmospheres*, *Environmental Science and Technology Air*, and *Atmospheric Chemistry and Physics*

AWARDS AND RECOGNITION

Julius F. Bartus Scholar, UT Chemical Engineering Fellowship, Scott Institute 2024 Seed Grant

FULL LIST OF PUBLICATIONS

1. D.M. Russell, F. Kunkler, J. Shen, M. Kohl, J. DeVivo, **N. Bhattacharyya**, C. Xenofontos, H. Klebach, L. Caudillo-Plath, M. Simon, E. Ahongshangbam, J. Almeida, A. Amorim, H. Beckmann, M. Busato, M.R. Canagaratna, A. Chassaing, R. Cruz-Simbron, L. Dada, P. Holzbeck, B. Judmaier, M. Kaniyodical Sebastian, P. Koemets, T. Krüger, L. Liu, M. Martinez, B. Mentler, A. Morawiec, A. Onnela, T. Petäjä, P. Rato, M. Reza, S. Ruhl, W. Scholz, E. Sommer, A. Tome, Y. Tong, J. Top, N. Silas Umo, G.R. Unfer, R.X. Ward, J. Weissbacher, B. Yang, W. Yu, M. Zauner-Wieczorek, I. Zgheib, J. Zhang, Z. Zheng, I. El Haddad, R.C. Flagan, A. Hansel, H. Junninen, M. Kulmala, K. Lehtipalo, J. Lelieveld, O. Möhler, S. Schobesberger, R. Volkamer, P.M. Winkler, D.R. Worsnop, T. Christoudias, A. Pozzer, N.M. Donahue, H. Harder, J. Kirkby, X-C He, and J. Curtius. Isoprene chemistry under upper-tropospheric conditions, *Nature Communications* (2025). <https://doi.org/10.1038/s41467-025-64229-w>
2. **N. Bhattacharyya**, B. Lopez, J. DeVivo, D.M. Russell, J. Shen, E. Sommer, J. Almeida, A. Amorim, H.M. Beckmann, M. Busato, M.R. Canagaratna, L. Caudillo, A. Chassaing, T. Christoudias, L. Dada, I. El-Haddad, R.C. Flagan, H. Harder, B. Judmaier, M. Kaniyodical Sebastian, J. Kirkby, H. Klebach, M. Kulmala, F. Kunkler, K. Lehtipalo, L. Liu, B. Mentler, O. Möhler, A. Morawiec, T. Petäjä, P. Rato, B. Rörup, S. Ruhl, W. Scholz, M. Simon, A. Tóme, Y. Tong, J. Top, N. Silas Umo, R. Volkamer, J. Weissbacher, D.R. Worsnop, C. Xenofontos, B. Yang, W. Yu, M. Zauner-Wieczorek, I. Zgheib, J. Zhang, Z. Zheng, X-C He, D. Stolzenburg, S. Schobesberger, J. Curtius, N.M. Donahue. Isoprene Aerosol Growth in the Upper Troposphere: Application of the Diagonal Volatility Basis Set to CLOUD Chamber Measurements. *Environmental Science and Technology Air* (2025). <https://doi.org/10.1021/acsestair.5c00106>
3. B. Lopez, **N. Bhattacharyya**, J. DeVivo, M. Wang, L. Caudillo-Plath, M. Surdu, F. Bianchi, Z. Brasseur, A., D. Chen, J. Duplissy, X-C He, V. Hofbauer, N. Mahfouz, V. Makhmutov, R. Marten, B. Mentler, M. Philippov, M. Schervish, D.S. Wang, S.K. Weber, A. Welti, I. El Haddad, K. Lehtipalo, M. Kulmala, D. Worsnop, J. Kirkby, R.L. Mauldin, D. Stolzenburg, S. Schobesberger, R. Flagan, N.M. Donahue. A diagonal volatility basis set to assess the condensation of organic vapors onto particles. *Environmental Science: Atmospheres* (2025). <https://doi.org/10.1039/D5EA00062A>
4. J. Shen, D.M. Russell, J. DeVivo, F. Kunkler, R. Baalbaki, B. Mentler, W. Scholz, W. Yu, L. Caudillo-Plath, E. Sommer, E. Ahongshangbam, D. Alfaouri, J. Almeida, A. Amorim, L.J. Beck, H. Beckmann, M. Bertheusel, **N. Bhattacharyya**, M.R. Canagaratna, A. Chassaing, R. Cruz-Simbron, L. Dada, J. Duplissy, H. Gordon, M. Granzin, L. Große Schulte, M. Heinritzi, S. Iyer, H. Klebach, T. Krüger, A. Kürten, M. Lampimäki, L. Liu, B. Lopez, M. Martinez, A. Morawiec, A. Onnela, M. Peltola, P. Rato, M. Reza, S. Richter, B. Rörup, M. Kaniyodical Sebastian, M. Simon, M. Surdu, K. Tamme, R.C. Thakur, A. Tomé, Y. Tong, J. Top, N. Umo, G. Unfer, L. Vettikkat, J. Weissbacher, C. Xenofontos, B. Yang, M. Zauner-Wieczorek, J. Zhang, Z. Zhang, U. Baltensperger, T. Christoudias, R.C. Flagan, I. El Haddad, H. Junninen, O. Möhler, I. Riipinen, U. Rohner, S. Schobesberger, R. Volkamer, P.M. Winkler, A. Hansel, K. Lehtipalo, N.M. Donahue, J. Lelieveld, H. Harder, M. Kulmala, D.R. Worsnop, J. Kirkby, J. Curtius, X-C He. New particle formation from isoprene under upper-tropospheric conditions. *Nature* (2024). <https://doi.org/10.1038/s41586-024-08196-0>
5. K.N. McPherson, L.G. Jahn, C.G. Masoud, **N. Bhattacharyya**, M. Modi, K. Patel, P. Abue, D.C. Blomdahl, P.K. Misztal, L. Hildebrandt Ruiz. Air pollution from unconventional oil and gas development in the Eagle Ford Shale. *Atmospheric Environment* (2024). <https://doi.org/10.1016/j.atmosenv.2024.120812>

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6. P. Abue, **N. Bhattacharyya**, M. Tang, D.C. Blomdahl, L.G. Jahn, D.T. Allen, R.L. Corsi, A. Novoselac, P.K. Misztal, L. Hildebrandt Ruiz. Emissions from hydrogen peroxide disinfection and their interaction with mask surfaces, *ACS Engineering AU* (2024). <https://doi.org/10.1021/acsengineeringau.3c00036>
7. C.G. Masoud, M. Modi, **N. Bhattacharyya**, L.G. Jahn, K.N. McPherson, P. Abue, K. Patel, D.T. Allen, L. Hildebrandt Ruiz. High Chlorine Concentrations in an Unconventional Oil and Gas Development Region and Impacts on Atmospheric Chemistry, *Environmental Science and Technology* (2023). <https://doi.org/10.1021/acs.est.3c04005>
8. L.G. Jahn, **N. Bhattacharyya**, D.C. Blomdahl, M. Tang, P. Abue, A. Novoselac, L. Hildebrandt Ruiz, P.K. Misztal. Influence of Application Method on Disinfectant Byproduct Formation during Indoor Bleach Cleaning: A Case Study on Phenol Chlorination, *Environmental Science and Technology Air* (2023). <https://doi.org/10.1021/acsestair.3c00011>
9. **N. Bhattacharyya**, M. Modi, L.G. Jahn, L. Hildebrandt Ruiz. Different Chlorine and Hydroxyl Radical Environments Impact *m*-xylene Oxidation Products, *Environmental Science: Atmospheres* (2023). <https://doi.org/10.1039/D3EA00024A>
10. **N. Bhattacharyya**, M. Tang, D.C. Blomdahl, L.G. Jahn, P. Abue, D.T. Allen, R.L. Corsi, A. Novoselac, P.K. Misztal, L. Hildebrandt Ruiz. Bleach Emissions Interact Substantially with Surgical and KN95 Mask Surfaces, *Environmental Science and Technology* (2023). <https://doi.org/10.1021/acs.est.2c07937>
11. A.M. Avery, M.W. Alton, M. Canagaratna, J.E. Krechmer, D.T. Sueper, **N. Bhattacharyya**, L. Hildebrandt Ruiz, W.H. Brune, A.T. Lambe. Comparison of the yield and chemical composition of secondary organic aerosol generated from the OH and Cl oxidation of decamethylcyclotrisiloxane, *ACS Earth and Space Chemistry* (2023). <https://doi.org/10.1021/acsearthspacechem.2c00304>
12. L.G. Jahn, M. Tang, D.C. Blomdahl, **N. Bhattacharyya**, P. Abue, A. Novoselac, L. Hildebrandt Ruiz, P.K. Misztal. Volatile organic compound (VOC) emissions from the usage of benzalkonium chloride and other disinfectants based on quaternary ammonium compounds. *Environmental Science: Atmospheres* (2023). <https://doi.org/10.1039/D2EA00054G>
13. A.T. Lambe, A.M. Avery, **N. Bhattacharyya**, D.S. Wang, M. Modi, C.G. Masoud, L. Hildebrandt Ruiz, W.H. Brune. Comparison of secondary organic aerosol generated from the oxidation of laboratory precursors by hydroxyl radicals, chlorine atoms, and bromine atoms in an oxidation flow reactor, *Environmental Science: Atmospheres* (2022). <https://doi.org/10.1039/d2ea00018k>
14. **N. Bhattacharyya**, A. Goodell, S. Rogers, A.H. Demond. Environmental impacts of wheat-based vodka production using life cycle assessment, *Journal of Cleaner Production* (2019). <https://doi.org/10.1016/j.jclepro.2019.05.226>
15. M. Otero-Diaz, B. Woo, **N. Bhattacharyya**, A.H. Demond. Enhancement of Aqueous Solubility of Organochlorine Pesticides by Ethanol, *Journal of Environmental Engineering* (2017). [https://doi.org/10.1061/\(ASCE\)EE.1943-7870.0001241](https://doi.org/10.1061/(ASCE)EE.1943-7870.0001241)

SELECTED PRESENTATIONS

1. N. Bhattacharyya, B. Lopez, J. DeVivo, A. Stinchfield, N.M. Donahue. Activity Measurements Constrain Anticipated Growth of Isoprene-Derived Particles in the Upper Troposphere. AGU Fall Meeting, Oral Presentation, Washington D.C., December 9-13, 2024.
2. **N. Bhattacharyya**, M. Tang, L.G. Jahn, D.C. Blomdahl, P. Abue, D.T. Allen, R.L. Corsi, A. Novoselac, P.K. Misztal, L. Hildebrandt Ruiz. Mask Prolong Exposure to Bleach Disinfection Byproducts. *AGU Fall Meeting, Oral Presentation*, December 12-16, 2022.
3. L. Hildebrandt Ruiz, **N. Bhattacharyya**, C.G. Masoud, C., K.N. McPherson, L.G. Jahn, P. Abue, M. Modi, K. Patel, D.C. Blomdahl, A.M. Avery, W.H. Brune, A.T. Lambe, P.K. Misztal. High Chlorine Emissions from Unconventional Oil and Gas Development Impact Atmospheric Composition through Radical Chemistry. *AGU Fall Meeting, Oral Presentation (Presenting Author)*, December 12-16, 2022
4. **N. Bhattacharyya**, P. Abue, L.G. Jahn, K. McPherson, E. Kang, W.H. Brune, P.K. Misztal, A.M. Avery, A.T. Lambe, L. Hildebrandt Ruiz. Cl Oxidation in Chamber and Flow Reactors: SOA Formation, Composition, and Gas Phase Chemistry, *AAAR, Oral Presentation*, Raleigh, NC, October 3-7, 2022.
5. **N. Bhattacharyya**, C.G. Masoud, K. McPherson, K. Patel, L.G. Jahn, P. Abue, D. Blomdahl, A.M. Avery, W.H. Brune, P.K. Misztal, A.T. Lambe, L. Hildebrandt Ruiz. Secondary Organic Aerosol Formation from *in situ* Cl

Oxidation of Ambient Air in an Oil and Gas Production Region, *AAAR*, Oral Presentation, Virtual, October 18-21, 2021.

6. **N. Bhattacharyya**, D. Blomdahl, L.G. Jahn, M. Tang, P. Abue, R.L. Corsi, D.T. Allen, A. Novoselac, P.K. Misztal, L. Hildebrandt Ruiz, Disinfection Impacts: Bleach Emissions and Interaction with Masks, *UT Austin COVID-19 Conference*, Poster Presentation, Virtual, November 10-11, 2020.
7. **N. Bhattacharyya**, M. Modi, L. Hildebrandt Ruiz, SOA Formation in Mixed Oxidant (OH + Cl) Environments, *AAAR*, Poster Presentation, Virtual, October 5-9, 2020.
8. **N. Bhattacharyya**, B. Woo, M. Otero Diaz, A.H. Demond, Co-solvent Effect on Partitioning of Fuel Compounds and Surface Water Contaminants, *Emerging Contaminants Summit*, Poster Presentation, Westminster, CO, March 1-2, 2016.
9. **N. Bhattacharyya**, J. Xiang, T. Tran, D. Chakravorty, Nature of Metal Ion Mediated Second Shell Hydrogen Bonds, *Southwest-Southeast American Chemical Society Regional Meeting*, Poster Presentation, Memphis, TN, November 4-5, 2015.

SKILLS & PROFICIENCIES

Instrumentation:	Gas and particle mass spectrometers
	FIGAERO-CIMS, NO ₃ ⁻ -CIMS, PTRMS, ACSM, AMS
	Particle size analyzers
	SEMS, SMPS
	Aerosol absorption and scattering
Experimental Designs:	Aethalometer, Nephelometer
	Gas phase monitors
	Ozone, NO _x , NO ₂ , SO ₂ , CH ₄ , formaldehyde
	Field measurements
	SE Texas, Taunus Observatory
Computer skills:	Laboratory Studies
	Oxidative flow reactor (OFR)
	Chamber experiments (Teflon, CLOUD)